

MCA-2 JAVA PROGRAMMING (2026)

Assignment -2

Unit 5: Generics & Common Classes

1. Explain generic class , interface methods with suitable example.
2. What are wildcards in Generics? Explain types with examples.
3. Differentiate between:
 - a. Unbounded wildcard (<?>)
 - b. Upper bounded (<? extends T>)
 - c. Lower bounded (<? super T>)
4. Differentiate String , StringBuffer and StringBuilder.
5. What are Wrapper classes? Explain autoboxing and unboxing.
6. Explain BigInteger, BigDecimal class with example.

Unit 6: Comparators & Lambda Expressions.

1. Differentiate between Comparable and Comparator.
2. What is a Lambda expression? Explain syntax and advantages.
3. Write a program using Lambda expression for sorting.
4. Explain method reference with types and examples.
5. Explain constructor reference with example.

Unit-8

1. Explain Collection Framework and its hierarchy.
2. What is **Iterable interface**? Explain its role.
3. Differentiate List vs Set vs Map
4. Explain java.time classes (LocalDate, ZonedDateTime)
5. Discuss Optional class and its role.

Unit-9

1. Discuss **IntSummaryStatistics**, **LongSummaryStatistics** and **DoubleSummaryStatistics**?
2. Explain commonly used methods of Collectors class:
 - toList()
 - groupingBy()
 - joining()
3. Explain Stream API in detail.
4. What is Collectors class? Explain its role.

Unit -11 Thread

1. Explain the **Thread class** in Java.
2. Explain thread methods:
 - a. start()
 - b. run()
 - c. sleep()

- d. `join()`
- 3. What are **thread priorities**? How are they set?
- 4. Explain **thread states** (New, Runnable, Running, Waiting, Terminated).
- 5. Write a short note on **inter-thread communication**.